A Hidden Markov Model (HMM) has been used to model states of human activity so far.

1. Background and motivation

Accelerometers and gyroscopes that are in smart phones identify what they're being used for, but noisy measurements obscure the true activity. My use case is a sensor-based, phone/software-only activity tracker – such as a fitness or wellbeing application that keeps track of the user's activity status (i.e. sitting, standing, walking or jumping) without the use of extra hardware – in order to measure and record sedentary time and active exercise without manual record keeping. Manual logging methods are unreliable and human activity occurs as sequences (people seldom change activity every fraction of a second): Obviously there needs to be human activity recognition, and there is no better way to model sequences than by using a Hidden Markov Model (HMM): hidden activity states, noisy sensor-derived observations, and transition probabilities that capture how activities are followed.

2. Collection and pre-processing of the data.

How often data is gathered and how it is gathered. All data were recorded by me on one iPhone 11 Pro with the Sensor Logger app (v1.68), logging the accelerometer (x, y, z) and gyroscope (x, y, z) every 10 ms. The native rate measured was 100.4 Hz (also known as median timestamp spacing). No sensor calibration was found in this device, so the uncalibrated sensor streams (accelerometer, gyroscope) were exported. Since there is only one phone, there was no need for cross-device harmonisation, and each recording was linearly interpolated on a common times scale of 50 Hz (also eliminating small time stamp jitter). 50 Hz is well above the requirement for walking and jumping (approx 2 Hz, Nyquist limit 25 Hz).

Protocol. With the phone at waist level (still: phone in a table) I made continuous takes for each activity. One jump, which was recorded as stationary (due to the phone being flat), was rejected (31.6 s). Seeking the needed training data and unseen test data is based on different takes (Table 1).

Activity	Training recording(s)	Unseen test recording	Clips (train / test)
Standing	63.0 s	51.7 s	9 / 7
Walking	67.3 s	65.9 s	10 / 9
Jumping	51.3 s	62.1 s	7 / 9
Still	53.1 s and 47.6 s	42.0 s	12 / 4
Total			38 / 29 clips (152 / 116 windows)

Cleaning. Timestamps were converted to seconds, sorted and de-duplicated. The first and last 3 s of every clip along with the last 6 s of each still clip (67 files in total) was removed to eliminate the hand motion of stopping and starting each clip, two clips of the phone were dropped and the rest of the clip was cut into labelled 6 s clips.

Windowing. Windows are 2 s long (100 samples at 50Hz) and 50% overlapping. Walking and jumping repeat with a frequency of about 1.5-2.5 Hz, thus a window of 2 s contains several cycles and provides a frequency resolution of 0.5 Hz; the number of observations is doubled with 50% overlap without losing information about the window boundaries.

Github link:https://github.com/bianca255/Formative-2---Hidden-Markov-Models

Video evidence.

Standing:https://drive.google.com/file/d/14uT6ARBWpLJfMW3SKGs14NQcADqROpiX/view?usp=sharing

Walking: https://drive.google.com/file/d/1UTu5RMnX_7XB9hBB2ldEmYN74YOK1OdT/view?usp=sharing

Jumping: https://drive.google.com/file/d/1kTEkOe85RHOPr82r3TRyutDD8_4DiyPr/view?usp=sharing

Still: https://drive.google.com/file/d/18V39JZcOXZ6BHVh_gU9oJ14RTGj7BDv4/view?usp=drive_link

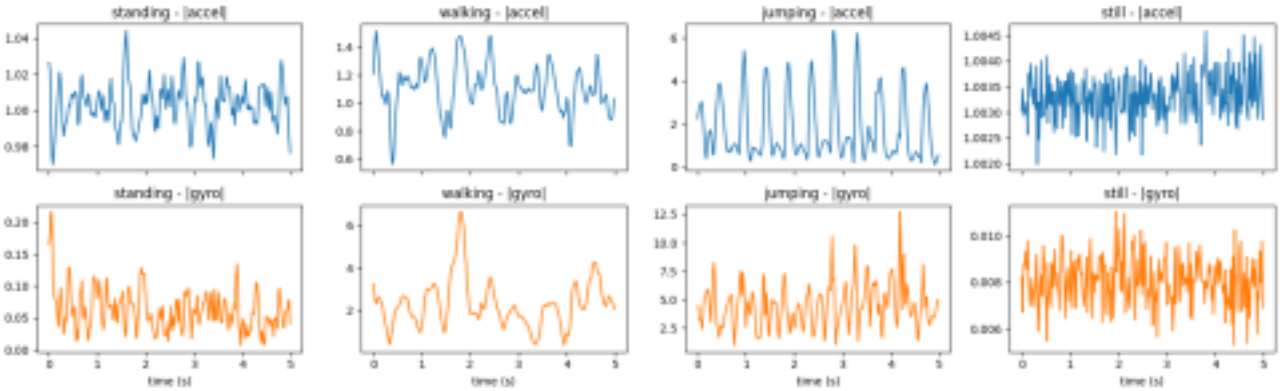


Figure 1. Example recording per activity: accelerometer magnitude and gyroscope magnitude . Note the different y-axis scales.

3. Feature extraction

One observation vector per window, 17 features per window: time domain – mean, standard deviation and RMS of the magnitudes of both accelerometer and gyroscope signals, the area of the signals in both magnitude signals (SMA) of both sensors, the correlation between accelerometer axes (xy, xz, and yz); frequency domain (FFT of the mean subtracted, Hann windowed signal magnitude) – dominant frequency, spectral energy and spectral entropy for both sensors. The measures of intensity of motion are standard deviation, RMS, SMA and energy; the measure of the cadence of motion is dominant frequency; the measure of motility being periodic (low entropy) or noise-like (high entropy) is spectral entropy; motion across the axes is correlated by axis correlation. All features are z-scored based on the mean and standard deviation of the training windows only instead of the test windows, since the Gaussian emission model requires all features to be on the same scale (this is often not the case, with the energy in the spectrum having an order of magnitude different from the frequency) and since the test windows could influence the scaling process.

Feature(s)	**Domain**	**Why it helps distinguish the activities**
Mean, std, RMS of |accel| and |gyro|	Time	Overall intensity and variability: jumping is far above walking, standing and still
SMA (accel, gyro)	Time	Total movement energy irrespective of direction; separates standing from walking
Correlation xy, xz, yz (accel)	Time	Jumping moves coherently across axes; resting phone axes are uncorrelated
Dominant frequency (accel, gyro)	Frequency (FFT)	Cadence of steps or jumps (about 1.7 Hz and 2 Hz); noise peaks for still
Spectral energy (accel, gyro)	Frequency (FFT)	Power of the periodic component; very high for jumping
Spectral entropy (accel, gyro)	Frequency (FFT)	Low for regular gait, high for noise-like signals such as a resting phone

4. HMM setup and implementation

Element	Definition in this project
Hidden states Z	Four activities: standing, walking, jumping, still
Observations X	17-dimensional z-scored feature vector per 2 s window
Transitions A	4x4 matrix, estimated from sequences of windows
Emissions B	Gaussian per state with diagonal covariance (mean and variance for each feature)
Initial pi	Probability that a sequence starts in each activity

The HMM is implemented from scratch, all in the log-domain because of numerical stability. The obvious choice here is vitterbi decoding, which gives the one most likely activity sequence — which is what is needed here, since the learned transitions respect vitterbi's path and sensitivity and specificity require an easy-to-decide label to be done every window.

structure. Baum-Welch (EM with the forward-backward algorithm) re-estimates pi, A and the Gaussian parameters to maximise the likelihood; training is stopped by a convergence check (or a maximum of 200 iterations if the total log-likelihood has not converged on convergence; variance floor 0.01, a relative change of the total log-likelihood below 1e-4). This check gets the likelihood also from the forward pass. Parameters are initialized from the labelled parts of the training windows, with each hidden state being a single activity, and a max voting over the training windows associating states with activity names. Every recording includes one activity and was stitched and run as a session of a single activity (one of each activity in random order; 36 training sessions) to allow learning of the transitions between activities.

5. Results and interpretation

Unseen data. The test set (29 clips, 116 windows) is not used in the training, nor in the normalisation and parameter fitting, but it consists of extracts from different recording takes recorded after the training takes. It is the same person, phone, location, tests generalisation between sessions, not people/phones. The test clips were demodulated by just one Viterbi decoder.

State (Activity)	Number of samples (windows)	Sensitivity	Specificity	Overall accuracy
Standing	28	1.00	1.00	1.00
Walking	36	1.00	1.00	1.00
Jumping	36	1.00	1.00	1.00
Still	16	1.00	1.00	1.00

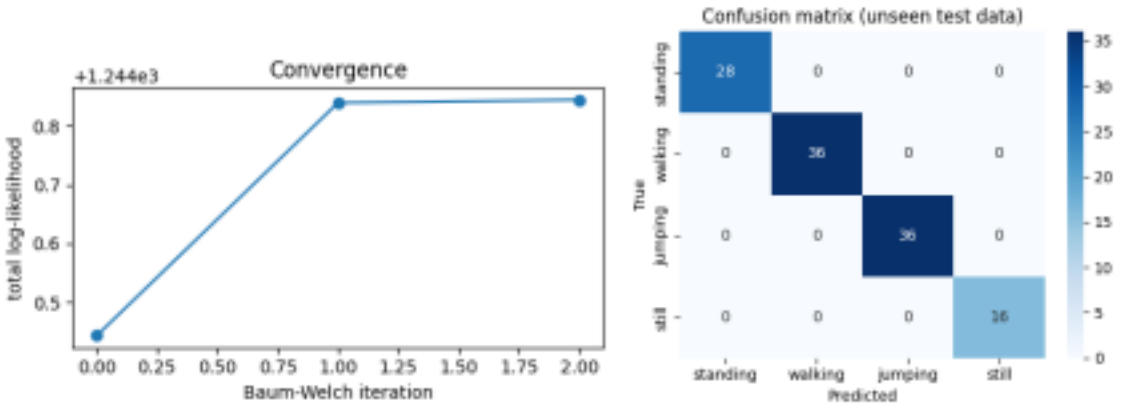


Figure 2. Left: confusion matrix on unseen data (all counts on the diagonal). Right: Baum-Welch log-likelihood per iteration (1244.44 to 1244.84 to 1244.85).

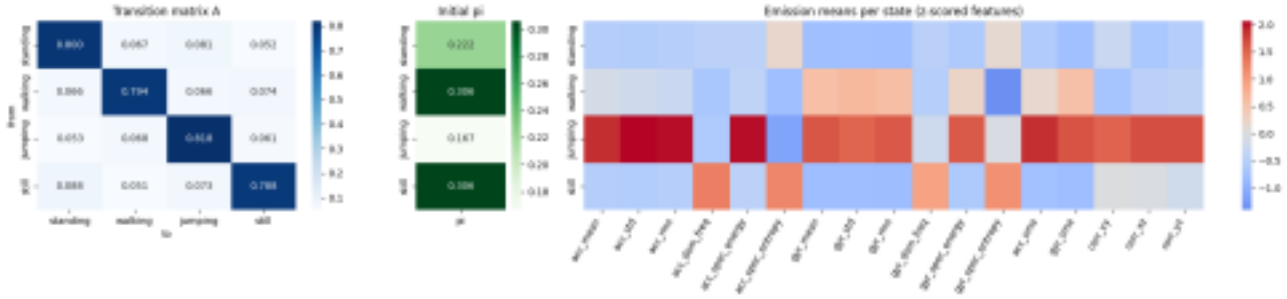


Figure 3. Learned transition matrix A, initial probabilities pi, and emission means per state (z-scored features).

Interpretation. The supervised initialisation was near to optimal (maximum likelihood), so the Baum-Welch did not return the data to this maximum likelihood solution, and it was terminated after three iterations. The probability of each activity staying in itself is 0.79–0.82, and the probability of each activity going to other activities is 0.05–0.09. The classes are separated by more than 1.5 standard deviations for accelerometer on spread and ‘jumping' vs. ‘rest' low intensity activity, ‘walking' moderate activity on gyroscope, ‘still' vs. ‘standing' low activity on energy and low spectral entropy (regular steps) on energy.

(noise-like).

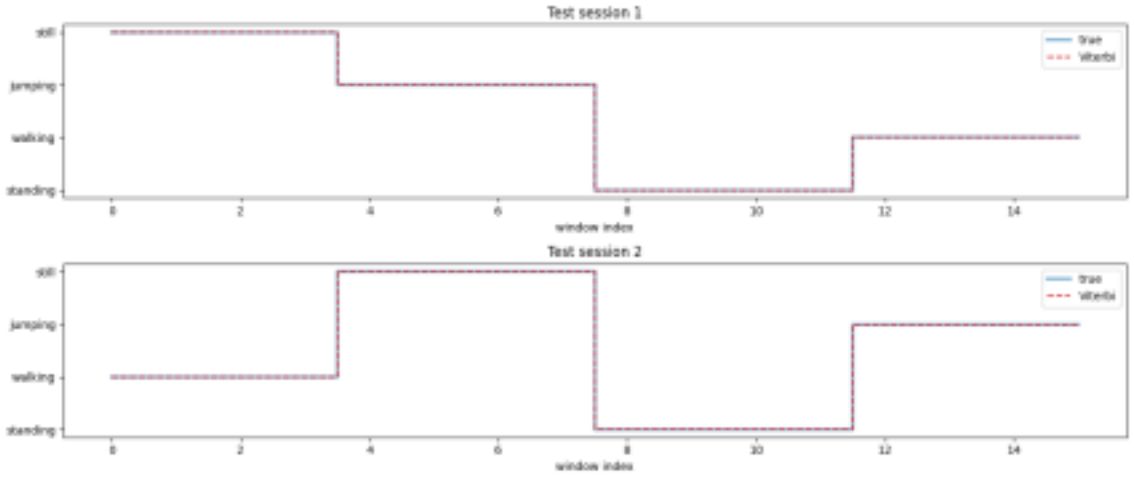


Figure 4. True activity (thick) versus Viterbi-decoded activity (dashed) for two unseen test sessions; the decoded path follows every activity change.

6. Discussion and conclusion

Most easy and most difficult tasks. All four activities were 100% accurate; degrees of difference are in margin.

The easiest is jumping; very large periodic accelerations every 2Hz. Standing and still are the two closest words; their amplitude features are nearly the same (accelerometer std approximately -0.57 and -0.59 z score units) they do differ on spectral features as well as the small tremor in the hands associated with standing. The step rhythm (regular for walking around 1.7 Hz) and gyroscope energy (when standing as opposed to walking) are separated.

Transition probabilities. The strong diagonal is realistic with activities that are not only occurring in time but are occurring in 'consecutive 2 s windows'. The off-diagonal are close to uniform as the images are recorded separately and stitched together rather than being representative of behaviour, in this case, a person will not likely start jumping straight to still position.

Noised and Noise rate. Noisy spectral features almost entirely come from the sensor noise, for the still recordings the standard deviation of the accelerometer is around 0.001 g and is very consistent. However, a resampling rate of 100Hz – 50Hz preserved all the relevant frequencies of motion (those frequencies below 5Hz), and using just one phone eliminated cross-device sampling differences.

Limitations and improvements. Both a participant and a phone has a completely different type of activity, and it's not something to be interpreted as general accuracy since a perfect score is possible. There is a small amount of training set (for instance, only 7 jumping clips) and test were recorded in the same setting. Improvements: (1) increase number of participants and phones, (2) more and longer recording of classes (e.g., sitting, running, and extra features, such as jerk/wavelets/zero-crossing rate etc.), (3) more and harder classes, e.g., full-covariance or Gaussian-mixture emission classes, and (4) pocket versus hand sensor placement test. Conclusion. An all-new Gaussian HMM with 17 features in the time and frequency domain was used to detect standing, walking, jumping and still, with unseen poses from the same individual, the learned parameters can be interpreted but more extensive data is needed before generalisation could be claimed.